

Data Quality & Exploration Report

Metadata Metric	Value / Count
Total Samples	49,475
Total Features	30
Memory Usage	11.70 MB
Duplicate Rows	1
Leakage Audit Verdict	WARN

Target Variable Class Balance

Class Label	Count	Percentage
Home Win (0)	24,244	49.00%
Draw (1)	11,253	22.74%
Away Win (2)	13,978	28.25%

Missing Values Summary

Feature Name	Missing Count	Missing %
home_fifa_rank	49,475	100.00%
away_fifa_rank	49,475	100.00%
fifa_rank_diff	49,475	100.00%
home_home_win_pct	327	0.66%
away_away_win_pct	321	0.65%
away_overall_win_pct	195	0.39%
away_draw_pct	195	0.39%
away_loss_pct	195	0.39%
... and 13 more		

Outlier Detection Audit

Outlier summary using IQR (interquartile range, threshold=1.5) and standard deviation Z-Scores ($|Z| > 3$). An unsupervised Isolation Forest flagged a global total of **2474** outliers across all numerical columns.

Feature	IQR Outliers	IQR %	Z-Score Outliers	Z-Score %
away_rest_days	4,968	10.04%	2,335	4.72%
home_rest_days	4,929	9.96%	2,311	4.67%
away_avg_goals_conceded	3,451	7.00%	727	1.48%
home_avg_goals_conceded	3,071	6.22%	715	1.45%
away_avg_goal_diff	2,728	5.54%	793	1.61%
away_draw_pct	2,551	5.18%	1,286	2.61%
home_avg_goal_diff	2,498	5.06%	744	1.51%
home_draw_pct	2,497	5.06%	1,124	2.28%
away_loss_pct	1,875	3.80%	635	1.29%
home_loss_pct	1,727	3.50%	636	1.29%
away_games_played	1,651	3.35%	806	1.64%
home_games_played	1,558	3.16%	775	1.57%
... showing top 12 of 30 numerical features.				

Data Leakage Checklist

Leakage audit check: **WARN**.

Flagged Column	Prohibited Keyword Match
home_avg_goals_scored	goals
home_avg_goals_scored	score
home_avg_goals_conceded	goals
away_avg_goals_scored	goals
away_avg_goals_scored	score
away_avg_goals_conceded	goals